PHY-112 | Principles of Physics-2

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Spring 2025 | Class #17

Department of Mathematics & Natural Sciences

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Uniting *E* and *B*: Electromagnetic Induction

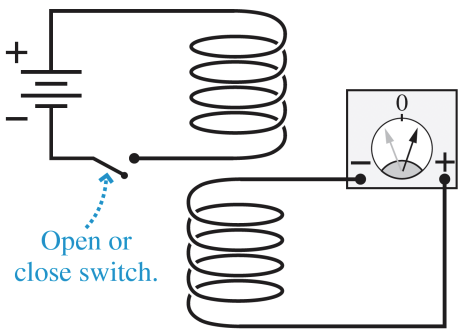
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Electromagnetic Induction: What is it?

The spark you see while unplugging a chord is due to induction

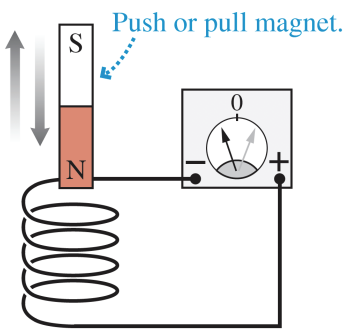
A magnetic field can create a current in a loop of wire without touching it. Only if the amount of field through the loop is changing. The process is called *Electromagnetic Induction*, and the current produced is called *Induced Current*, first discovered by Michael Faraday in 1831.

Experiment #1



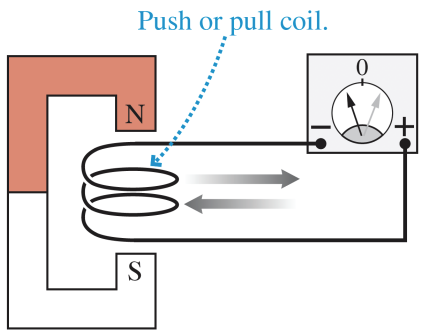
Opening or closing the switch creates a momentary current.

Experiment #2



Pushing the magnet into the coil or pulling it out creates a momentary current.

Experiment #3



Pushing the coil into the magnet or pulling it out creates a momentary current.

Faraday's Law of Induction

This is the 3RD Maxwell's Equation

An EMF $\mathcal{E}$ is induced around a closed loop if the magnetic flux through the loop changes. The magnitude of the EMF is

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \text{ For One-turn coil}$$

$$= -N\frac{d\Phi_B}{dt} \text{ For } N\text{-turn coil}$$

and the direction of the EMF is such as to oppose the flux change.

The EMF associated with a changing magnetic flux, regardless of what causes the change, is called an induced EMF $\mathcal{E}$. Then, if there is a complete circuit having resistance R, a current

$$I_{\text{induced}} = \frac{\mathcal{E}}{R}.$$

How to vary this Magnetic Flux?

Each of this causes induction

How many ways can You change the Magnetic Flux?

- By changing the current through a current loop
- By bringing a permanent magnet closer to a current loop/coil
- By bringing the current loop/coil closer to a permanent magnet

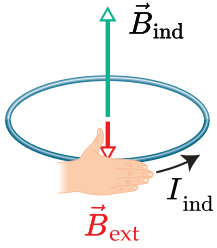
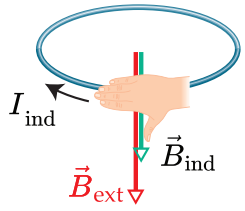
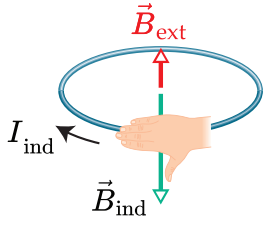
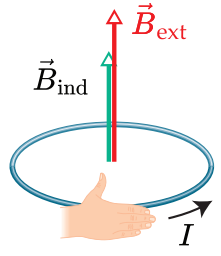
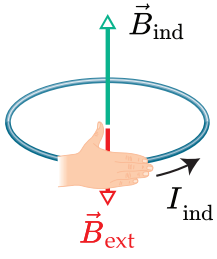
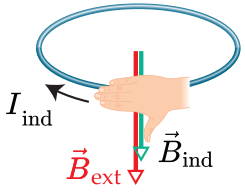
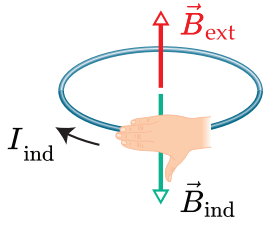
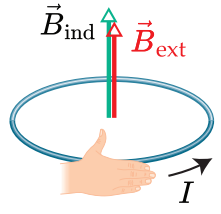
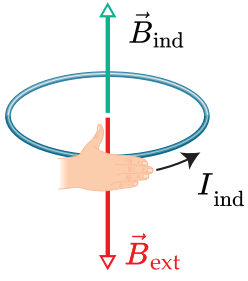
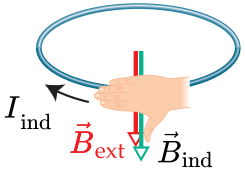
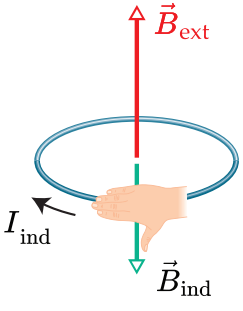
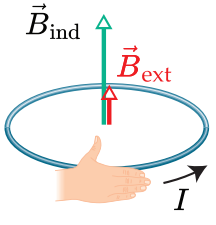
Incepting Ideas (1)

Hint: Discretize the Flux expression

Q: A 750-turn coil of wire 2.5 cm in diameter is in a magnetic field that increases from 0.20 T to 0.50 T in 15 ms. The coil has a resistance of the order 2 Ω. The axis of the coil is parallel to the field. (i) What is the induced EMF of the coil? (ii) What is the induced current in the coil?

Lenz's Law © Veritasium

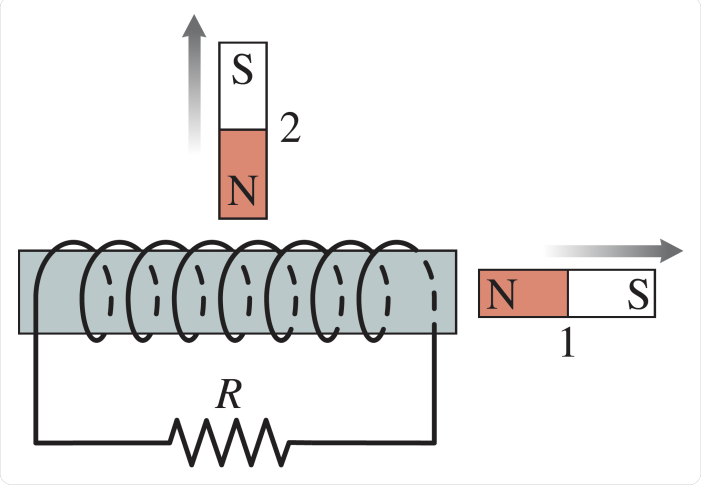
Oh, No! You don't, Law!

	Increasing the external field induces a current with an induced field that *opposes the change*.	Decreasing the external field induces a current with an induced field that *opposes the change*.	Increasing the external field induces a current with an induced field that *opposes the change*.	Decreasing the external field induces a current with an induced field that *opposes the change*.
The induced current creates this field, trying to offset the change.				
The fingers are in the current's direction; the thumb is in the induced field's direction.				
				

Incepting Ideas (2)

Hint: Check the main motion and oppose it

Q: A solenoid is wound as shown. (i) Is there an induced current as magnet 1 is moved away from the solenoid? If so, what is the current direction through resistor *R*?



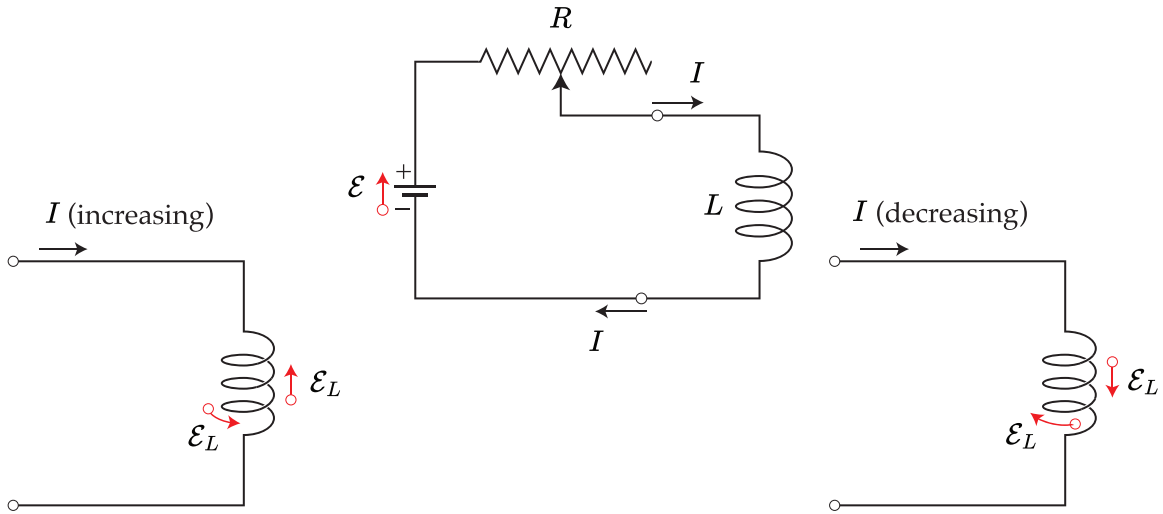
(ii) Is there an induced current as magnet 2 is moved away from the solenoid? If so, what is the current direction through resistor *R*?

Inductors and Inductance

Inductance opposes change in current

The induced magnetic flux Φ_B (One unit of it per turn of the coil) is proportional to the current i in the coil.

$$L = \frac{N\Phi_B}{i}.$$



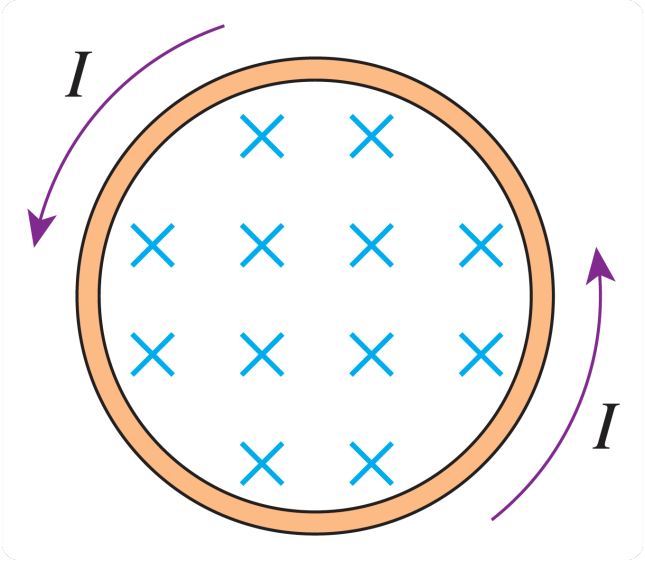
The induced EMF $\mathcal{E}$ can now be defined using Faraday's Law of induction.

$$\mathcal{E}_L = -\frac{d(N\Phi_B)}{dt} = -L\frac{di}{dt}.$$

Incepting Ideas (3)

Hint: Check the main motion and oppose it

Q: There is a *clockwise* induced current in the conducting loop shown. Is the magnetic field inside the loop increasing in strength, decreasing in strength, or steady?



STOP! NEW PHYSICS!

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What Does Faraday's Law Tell Us?

Two parters: Motional and Inducive EMF

Faraday recognized that all induced currents are associated with a changing magnetic flux. There are two fundamentally different ways to change the magnetic flux through a conducting loop:

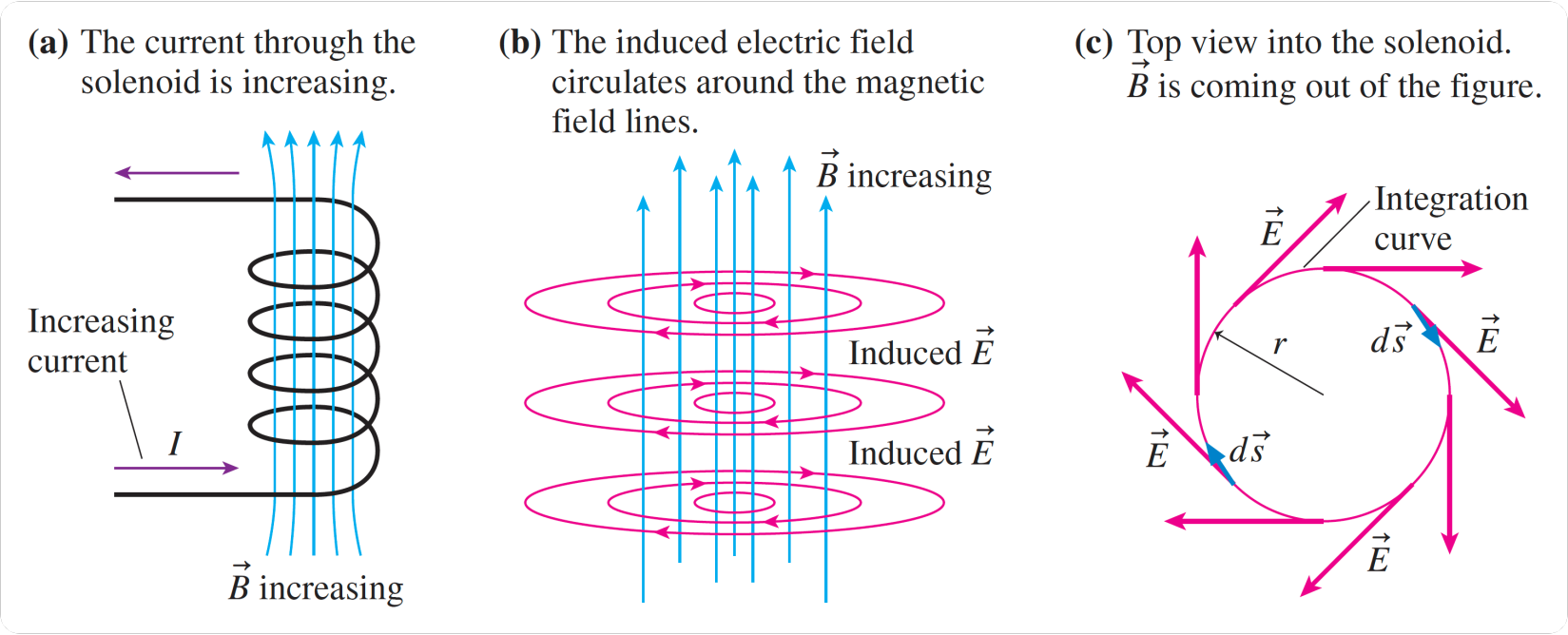
- The loop can expand, contract, or rotate
- The magnetic field can change

$$\mathcal{E} = \left|\frac{d\Phi_B}{dt}\right| = \left|\frac{d\vec{B}\cdot\vec{A}}{dt}\right| = \left|\vec{B}\cdot\frac{d\vec{A}}{dt} + \vec{A}\cdot\frac{d\vec{B}}{dt}\right|$$

The first term on the right side represents a **Motional EMF**. The second term on the right side is the **NEW Physics!!!** in Faraday's law. It says that an EMF can also be created simply by changing a magnetic field, even if nothing is moving.

Where is this $\vec{E}$ field?

This electric field has a curl



Note: Circular $\vec{E}$-field lines violate the rule that $\vec{E}$-field lines have to start and stop on charges. However, that rule applied only to Coulomb fields created by source charges. An induced $\vec{E}$-field is a non-Coulomb field created not by source charges but by a changing magnetic field. Without source charges, induced $\vec{E}$-field lines must form closed loops.

The 3rd Maxwell's Equation

Magnetic field now produces an Electric field

An alternative statement of Faraday's law relates the induced $\vec{E}$-field to the changing $\vec{B}$-field.

$$\mathcal{E} = -\oint \vec{E} \cdot d\vec{r} = \vec{A} \cdot \frac{d\vec{B}}{dt}$$

$$\oint \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A} \quad \textbf{(Integral Version)}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \textbf{(Differential Version)}$$

The induced electric field is caused not by charges but by a changing magnetic field. It is called a **non-Coulomb $\vec{E}$-field**.

That is it for this semester!

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